I have one of the original versions of the SteppIR DB36/80 that I acquired through an estate sale in 2018. By design or coincidence, the SK was a retired career firefighter, as am I. The best guess is that it's circa 2008 vintage. There have been several modifications and changes to the DB36/80 design since then.

The DB36/80 is unique in some ways (the reflector is the only one of the SteppIR beam design to not put the EHU placement at the end of the boom; it has fewer relays than a DB42 and only one driven element for 40-6M, the only relay is to use 60/80 meters on a dipole with the reflector and director 2 [D2] as capacity hats).

After some years of Nevada solar abuse before I got it then more years here in North Idaho, it was time for a major rehab. Some of the fiberglass fibers of the tubes were showing and it needed some TLC as well as fixing some issues that had come up:

- no longer tuning on 60/80M
- the RX/TX pattern was definitely not what was normal or expected
- the gain was lower than previous
- Front to back ratio was well below norms
- 40M tuning when below 35 F was not possible (but was below 25F)
- Tuning to the bottom of 40M was not possible
- The noise floor was raised significantly higher (average of 20 dB)

Once the tower was tilted and the DB36/80 was broken down, it became clear that *EVERY* problem that had been presented over the years to SteppIR antennas in general, was present here. It was WELL past time to do a complete rehab/repair.

Reflector tape (which I had replaced a few years ago) showed some accordion at the ends. That was problem enough to note that the rest of the tape was ignored.

Driven, smelled like something had gotten too hot (still functions but I had replaced it two years ago, which also meant a change of mounting plate). That was likely because to snag a few 40M DX slots, I used the tuner and kept the power down (don't do that!).

[Moral of the previous driven EHU: DO NOT run more than 1 KW on 6M, it's beyond the un-un/balun design (to work at all on 6M). It quit, just as I closed the DX contact, an expensive lesson. One type balun will NOT correctly cover all HF, let alone 80-6M.]

Director 1 (20M passive) had case cracks (original old style case).

Director 2 had a broken weld on one of the sprockets, which caused excessive spacing from the tapes, which were badly scarred with the holes elongated. This explains the lack of tuning on 60/80 (it's a cap hat), the poor pattern and lack of gain (down at least 3 dB). Fortunately, the tapes had retracted or it would have been a tedious manual retract.

ALL EHU showed signs of copper dusting, none serious.

Decisions: Replace ALL of the EHU, 2 of the mounting plates (because the EHU style had changed); new mounting hardware for all of the new EHU and a complete rewire from the control box on the antenna, to each EHU (because now they come with plugs and cutting off the old wires made them too short).

Added repairs:

- Replace the entire tower run of controller wire (into a utility box at the base of the tower, the rest of the run is in conduit to the shack).
- Replace the rotator control cable (to the same utility box, different conduit run).
- Lube all bearings (tilt plate and thrust bearing)

When I purchased the antenna, it was taken off of the tower (which I also bought, with a prop pitch rotor), the tubes with sweeps removed and the boom broken into sections. It was all placed on a trailer, hauled here and reassembled, put back on the air.

I had not gone through the assembly process more than this. Was I *ever* in for a learning period. I hoped for a week max of off air time (but did work some DX with my wire antennas; ugly but it got done). I got schooled.

With the help of Tom, KJ7TEA (the only other DXer in my county); we proceeded to get this done and it took ten days. Ten LONG, TEDIOUS, PHYSICAL tiring days, some were 12 hour days. Tom was patient and understanding at my ignorance since he had just put new sweeps on his DB42/80 and understood the process. He's also a former radio tech and understood "Do it right, or don't bother" (so he took care of the wiring and trivia). As a firefighter, I leaned towards shortcuts but not the long game (git 'er dun!).

Tasks that were presented:

- Sand and paint each tube (including extensions, some showing loose fiber hairs)
- Attach new sweeps (in detail later)
    - With the new ends on each tube
    - Epoxy to bind the new ends to the tube
    - With the heat shrink to seal the joints
    - Then PEX clamps and new clamps with spacing rods
    - Scotch 88 tape (NOT 33+) EVERYTHING at EVERY joint
- Fully extend each tube
    - because the original builder did not but shrunk them to almost the proper length, which allowed them (taped only) to slip even shorter
    - three fully extended tubes were short (and not one was ever cut, the original end grooves still in place, I pulled but didn't 'gorilla' them)
    - cut to the proper length (the 3 shortest [by an inch on one] tubes placed on the D1 passive and matched to one side of D2 so there was no variation in length on any side pair)
- Protect ALL exposed wire on the antenna
    - placed in curly plastic wrap designed for that purpose
    - tie wrapped to the <u>under</u> side of the boom (no bird talon damage allowed, including the bald eagles here, added UV protection)

    - More Scotch 88 over each tie wrap to the boom (UV protection)
- Tidy the cables from the tower top to the control box allowing slack for rotor turn
- Replace the UV eaten tower run of rotator and controller cable to the utility box at the base of the tower (the rest of the runs are underground to the shack in separate conduit from the coax, a third conduit provides AC power at the tower)
- **ALL** wires connected to static protection at the utility box at the tower base
- Tidy the wiring disaster in the box (a bowl of wire spaghetti); all wiring to professional standards. (I didn't take a picture of the before, it was UGLY.)

The 'latest' version on the SteppIR web site of how to attach the sweeps to the tubes is over a year and half old while discussion in this group leaves a lot of questions to the best methods. I offer this as one way to attach the sweeps; time will tell if it's the best way.

For these tasks, we used

- Marine grade 2 part epoxy primer then paint over all tubes (extensions too, all of it). This is outdoor only, the fumes are super nasty (and wear throw away clothes)
- a two part epoxy to add the new ends that join tubes to sweeps (waited for it to set, which happened to be the next day because it started raining)
- heat shrink tubing (part of the new sweep kit) over the sweep joints
- heat shrink over EACH tube joint (the belt)
- tape over ALL of the heat shrink even though it's UV rated (long days are high UV here plus suspenders for the belt)
- two PEX clamps over the joint, sweep side (a firm belief in belt AND suspenders)
- tape **again**, over the entire joint, all parts covered (for added UV resistance)
- install the kit spacing rod clamps at the END of the sweep heat shrink (securing the other end of the joint per the modern design, being VERY CAREFUL to not pinch OR cut the tape!). In this case, the tube was a little too thin, so a layer or two of tape wrap was used to shim a tight fit.

Here are some (attached in email form) pictures of how we attached the sweep to the tubes to also help the visual learners.

The next issue that rose up (not photographed) is the new style of truss rope placement, which needed a new additional clamp for the tubes. Rather than wait for (another) shipping period (SteppIR takes 2-3 days to ship normally), we

- took the old rod clamps off of the old sweeps
- cut the barbed ends off (that went into the old sweeps)
- enlarged the older eye bolt hole to connect the new rope directly
- set the new clamps 5" from the second (from the sweep end) joint, sweep side

Once the tubes with sweeps were in place, we then ran new (heavier, what we had on hand) antenna rope from one clamp through the original turnbuckle at the truss support (a thimble in place for after gross set up so there was no rope to metal movement) to the opposite clamp (per the new design).

We carefully balanced the rope tension to match the Director 1 element sag (it has no sweep, a baseline for sag) WITHOUT bowing out/up the tubes (KEY POINT, that's TOO TIGHT, forget level, it's impossible). Set final element twist by rotating the tubes (I look at the end rods compared to the boom for 'level') then clamp the ropes at the thimble so they are locked in place (no rope slips).

Then final twists removed by rotating the tubes if required, when done tighten their (Fernco) clamps.

Then testing (resistance, opens, motor) per the guidelines (all passed with flying colors); retest the motors by gross resonance checks at EACH band with an antenna analyzer while still low to the ground. All passed, no issues noted, no odd noises except a HUGE sighs of relief from both of us.

Tape ALL of the Fernco rubber adapters (two wraps each, Scotch 88). NO EXPOSED RUBBER ANYWHERE, the sun will destroy it.

A final check of all fittings (tightness, butyl the opening to the control box, tape and butyl all RF connectors) and tilt the tower back to vertical (a COMPLETE upper body workout!). Test the rotor motion (newly cabled), then raise the tower.

Suggested: Reset ALL former SDA100/2000) settings AND START OVER, factory fresh, then reworking the adjusting on ALL bands, as if it were new (well, it IS in this case, only the boom was untouched).

As it happened, the next day the high bands were SMOKING hot, so everything (once retuned) got a full test, at full legal limit (SWR better than ever!) and comparisons were made to Tom's DB42. That was a good day for two very exhausted DXers.

New band mode slots (collect the entire set!) were acquired, front to back testing was done and it appears that this was a complete successful recovery for the DB36/80. The noise floor even fell to the previous nominal.

While the dipole tuning settings are not completed yet (it takes time and we're both still pretty exhausted at this point), I score this as a huge win; ALL of the issues noted (including tuning to the bottom of 40M) are resolved. I'll complete the 60/80 tuning task upon MY recovery, because low band season is about to begin. (I have an 80M ground plane and 160L antennas too; the trees are convenient masts.)

The final test will happen this winter to see if it works when cooler (below 35F). I expect that will be successful also with the new sweeps and corrected issues (and 36V power supply). Fingers crossed!

It took FAR longer for two guys in their 60's (mentally younger of course!) than I had ignorantly anticipated and would have taken WEEKS without the help and advice of Tom, KJ7TEA. His help was instrumental in finding and resolving the issues along the way of this journey. It was his help that allowed this to be finished before Fall got too wet or Winter snows arrived stopping any hope of completion.

Now, back to the hunt for DX and the never ending task of reducing sources of noise to lower the floor to MDS; may THAT day be soon (Tom is already there).

73,
Rick nk7i